

Q1 Transpose of a square matrix.

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9

 $\Rightarrow$

	0	1	2
0	1	4	7
1	2	5	8
2	3	6	9

$(0,1) \rightarrow (1,0)$
 $(2,0) \rightarrow (0,2)$

$(1,1) \rightarrow (1,1)$

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9

```
for (int i = 0 ; i < n ; i++) {
```

```
    for (int j = i+1 ; j < n ; j++) {
```

```
        int temp = arr[i][j]
        arr[i][j] = arr[j][i];
        arr[j][i] = temp;
```

```
    }
```

```
}
```

Q2 Rotate a square matrix by 90° .

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9

$\Rightarrow$

	0	1	2
0	7	4	1
1	8	5	2
2	9	6	3

↓ transpose

	0	1	2
0	1	4	7
1	2	5	8
2	3	6	9

reverse
each
r

	0	1	2
0	7	4	1
1	8	5	2
2	9	6	3

Todo!

Problem Statement

Given an array of 1's and 0's, you are allowed to replace only one 0 with 1. Find the maximum number of consecutive 1's that can be obtained after making the replacement.

Note: This question has been asked in many companies like *Amazon, Microsoft, Adobe*, etc.

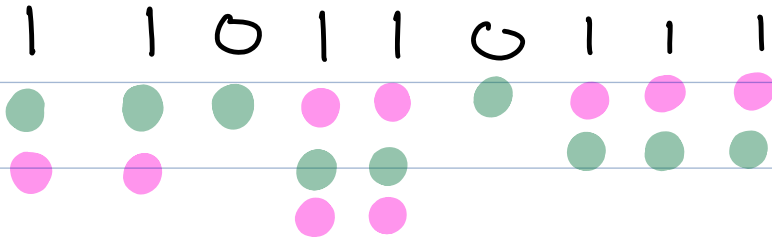
Example 1

Input = [1, 1, 0, 1, 1, 0, 1, 1]

Output = 5

Explanation:

If we replace 0 at 2nd index or 0 at 5th index with 1, in both cases we get 5 consecutive 1's.



int ans = 0;

for (int i = 0; i < n; i++) {

if (arr[i] == 0) {

l = 0, r = 0;

// left

int j = i - 1;

while (j >= 0 && arr[j] == 1)

l++, j--;

// right

j = i + 1;

while (j < n && arr[j] == 1)

r++, j++;

ans = max(ans, l + r + 1);

}

}

SUMMARY

CASE 99



all ones

111111

0 1 1 1 0 1 1 1 0


1 1 1 1 0 1 1 1 1


max iteration
per element $\Rightarrow 3$

Total iterations $\Rightarrow 3n$

$T_c : O(n)$

1 1 1 1 1 1 0


```
int flag = 1
```

```
for (int i = 0; i < n; i++) {
```

```
    if (arr[i] == 0) {
```

```
        flag = 0, break;
```

```
    }
```

```
if (flag == 1)
```

```
    return n;
```

```
int ans = 0;
```

```
for (int i = 0; i < n; i++) {
```

```
    if (arr[i] == 0) {
```

```
        l = 0, r = 0;
```

```
        // left
```

```
        int j = i - 1;
```

```
        while (j >= 0 && arr[j] == 1)
```

```
            l++, j--;
```

```
        // right
```

```
        j = i + 1;
```

```
        while (j < n && arr[j] == 1)
```

```
            r++, j++;
```

```
        ans = max(ans, l + r + 1);
```

```
    }
```

```
return ans;
```

Problem Statement

Given an array of 1's and 0's, find the maximum number of consecutive 1's that can be obtained by SWAPPING at most one 0 with 1 (already present in the string).

Note: This question has been asked in many companies like *Amazon*, *Direct-i*, *Microsoft*, *Adobe*, etc.

Example 1

Input: [1, 0, 1, 1, 0, 1]

Output: 4

Σ_{x1} 1 0 1 1 0 1
 ¹ ~~0~~ ⁰

Σ_{x2} 1 1 1 0 1 1
 ¹ ~~0~~ ⁰

Σ_{x3} 1 1 1 0 1 1 0 0 1

0
~~1~~ 1 0 1 1 ~~0~~ 1 1 1 1
 2 4

2 + 4 + 1 = 7

7

```
int total_ones = 0
```

```
for (int i = 0; i < n; i++) {
```

```
    if (arr[i] == 1)
```

```
        total_ones++;
```

```
}
```

```
if (total_ones == n)
    return n;
```

```
int ans = 0;
```

```
for (int i = 0; i < n; i++) {
```

```
    if (arr[i] == 0) {
```

```
        l = 0, r = 0;
```

```
        // left
```

```
        int j = i - 1;
```

```
        while (j >= 0 && arr[j] == 1)
```

```
            l++, j--;
```

```
        // right
```

```
        j = i + 1;
```

```
        while (j < n && arr[j] == 1)
```

```
            r++, j++;
```

```
        if (total_ones > (l + r))
```

```
            ans = max(ans, l + r + 1);
```

```
    } else
```

```
        ans = max(ans, l + r);
```

```
}
```

1 1 1 0 1 1 1

$$\text{total_ones} = 6$$

$$l = 3$$

$$r = 3$$

$$l + r$$

1 1 1 0 1 1 1 0 1 0 1

$$\text{total_ones} = 8$$

$$l = 3$$

$$r = 3$$

$$l + r + 1$$

Q Reverse String word by word.

S = "scaler is the best"

Ans = "best the is scaler"

S = "scaler is the best"
0 1 2 3 4 5 6 7 8 9 10 11

⇓ reverse entire string

ts eb ch t si xelac s

⇓ reverse word by word.

best the is scaler